MSG No. 71.535 $Im'mm$ [Type III, orthorhombic]

Table 1: Wyckoff site: 2a, site symmetry: m'mm

No.	position	mapping
1	$[0, 0, 0]$	[1,2,3,4,5,6,7,8]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[9,10,11,12,13,14,15,16]

Table 2: Wyckoff site: 2b, site symmetry: m'mm

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8]
2	$\left[\frac{1}{2}, 0, 0\right]$	[9,10,11,12,13,14,15,16]

Table 3: Wyckoff site: 2c, site symmetry: m'mm

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[1,2,3,4,5,6,7,8]
2	$\left[0, 0, \frac{1}{2}\right]$	[9,10,11,12,13,14,15,16]

Table 4: Wyckoff site: 2d, site symmetry: m'mm

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8]
2	$\left[0, \frac{1}{2}, 0\right]$	[9,10,11,12,13,14,15,16]

Table 5: Wyckoff site: 4e, site symmetry: 2mm

No.	position	mapping
1	$[x, 0, 0]$	[1,2,3,4]
2	$[-x, 0, 0]$	[5,6,7,8]
3	$\left[x + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[9,10,11,12]
4	$\left[\frac{1}{2} - x, \frac{1}{2}, \frac{1}{2}\right]$	[13,14,15,16]

Table 6: Wyckoff site: 4f, site symmetry: 2mm

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,2,3,4]
2	$\left[-x, \frac{1}{2}, 0\right]$	[5,6,7,8]
3	$\left[x + \frac{1}{2}, 0, \frac{1}{2}\right]$	[9,10,11,12]
4	$\left[\frac{1}{2} - x, 0, \frac{1}{2}\right]$	[13,14,15,16]

Table 7: Wyckoff site: `4g`, site symmetry: `m'2'm`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,4,5,8]`
2	$[0,\ -y,\ 0]$	`[2,3,6,7]`
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,12,13,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[10,11,14,15]`

Table 8: Wyckoff site: `4h`, site symmetry: `m'2'm`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,4,5,8]`
2	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[2,3,6,7]`
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[9,12,13,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[10,11,14,15]`

Table 9: Wyckoff site: `4i`, site symmetry: `m'm2'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,3,6,8]`
2	$[0,\ 0,\ -z]$	`[2,4,5,7]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,11,14,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,12,13,15]`

Table 10: Wyckoff site: `4j`, site symmetry: `m'm2'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,3,6,8]`
2	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[2,4,5,7]`
3	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,11,14,16]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,12,13,15]`

Table 11: Wyckoff site: `8k`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,15]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[2,16]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,13]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[4,14]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[5,11]`
6	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[6,12]`
7	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[7,9]`
8	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[8,10]`

Table 12: Wyckoff site: `8l`, site symmetry: `m'..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,8]`
2	$[0,\ -y,\ -z]$	`[2,7]`
3	$[0,\ -y,\ z]$	`[3,6]`
4	$[0,\ y,\ -z]$	`[4,5]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,16]`
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[10,15]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[11,14]`
8	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[12,13]`

Table 13: Wyckoff site: `8m`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,3]`
2	$[x,\ 0,\ -z]$	`[2,4]`
3	$[-x,\ 0,\ -z]$	`[5,7]`
4	$[-x,\ 0,\ z]$	`[6,8]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,11]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,12]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[13,15]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[14,16]`

Table 14: Wyckoff site: `8n`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,4]`
2	$[x,\ -y,\ 0]$	`[2,3]`
3	$[-x,\ y,\ 0]$	`[5,8]`
4	$[-x,\ -y,\ 0]$	`[6,7]`
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,12]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[10,11]`
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[13,16]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[14,15]`

Table 15: Wyckoff site: `16o`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[x,\ -y,\ -z]$	`[2]`
3	$[x,\ -y,\ z]$	`[3]`
4	$[x,\ y,\ -z]$	`[4]`
5	$[-x,\ y,\ -z]$	`[5]`
6	$[-x,\ -y,\ z]$	`[6]`
7	$[-x,\ -y,\ -z]$	`[7]`

continued ...

Table 15

No.	position	mapping
8	$[-x,\, y,\, z]$	[8]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[10]
11	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[11]
12	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[12]
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[15]
16	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[16]